

COS760 Project – Group 16

Semantic Relatedness Using SemRel Datasets: Evaluating Cross-Lingual Transfer and Data Augmentation for Semantic Relatedness in Afrikaans, Hausa and Kinyarwanda

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Abstract

We investigate semantic relatedness in three African languages – Afrikaans, Hausa, and Kinyarwanda – using the SemRel dataset (Abdulmumin et al., 2024). We evaluate three modelling approaches: an unsupervised cosine similarity baseline, cross-lingual transfer from XLM-R (Conneau et al., 2020), and fine-tuning of the African-focused AfriBERTa model (Alabi et al., 2022). We additionally apply back-translation data augmentation to the two lowest-resource languages. Our results show that XLM-R zero-shot transfer achieves strong performance for Afrikaans ($\rho = 0.818$), reflecting its linguistic proximity to English, while Hausa and Kinyarwanda benefit substantially from language-specific fine-tuning (+0.044 and +0.152 respectively). Contrary to our hypothesis, XLM-R consistently outperforms AfriBERTa across all languages and conditions, suggesting that scale of pretraining data outweighs language-specific focus for this task at the model sizes evaluated. Data augmentation via back-translation provides modest improvements for both low-resource languages (XLM-R: +0.012 Hausa, +0.009 Kinyarwanda), constrained by the quality of available multilingual translation models for Hausa.

1 Introduction

Semantic relatedness is a fundamental task in natural language processing (NLP) that measures the degree of meaningful connection between two pieces of text. Unlike semantic similarity, which captures only paraphrase-like relationships, semantic relatedness encompasses a broader range of associations, including topical, functional, and causal connections. This distinction is important for downstream applications such as information retrieval, question answering, machine translation, and dialogue systems.

Significant progress has been made in semantic relatedness for high-resource languages such as En-

glish, largely due to the availability of large annotated datasets and powerful pretrained transformer models. However, performance for low-resource languages – particularly African languages – remains substantially limited. The African continent is home to over 2,000 languages spanning several distinct language families, yet the vast majority are dramatically underrepresented in existing NLP benchmarks and pretrained model corpora.

This project investigates semantic relatedness in three African languages – Afrikaans, Hausa, and Kinyarwanda – using the SemRel dataset (Abdulmumin et al., 2024). These languages use the Latin script while representing distinct language families: Afrikaans is a West Germanic language with strong structural similarity to English; Hausa is a Chadic language spoken across West Africa; and Kinyarwanda is a Bantu language spoken primarily in Rwanda. This diversity allows for meaningful analysis of how linguistic distance affects cross-lingual transfer.

The central research questions are: (1) How effectively can models trained on English transfer to Afrikaans, Hausa, and Kinyarwanda in a zero-shot setting? (2) How do general multilingual models compare to African-focused models in capturing semantic relatedness? (3) To what extent does back-translation augmentation improve performance for low-resource languages?

Problem Statement. Despite recent advances in multilingual NLP, semantic relatedness modelling for low-resource African languages remains significantly underperforming compared to high-resource languages. The core problem is threefold: (1) existing multilingual models are pretrained on data that severely underrepresents African languages, limiting their zero-shot transfer effectiveness; (2) African-focused models exist but their advantage over general multilingual models for specific downstream tasks remains empirically unverified; and (3) data augmentation strategies for African languages

are constrained by the poor quality of available machine translation resources. This study directly addresses these gaps by providing systematic empirical evidence across three typologically distinct African languages using a standardised benchmark dataset.

Responsible NLP: This work is motivated by the goal of more equitable NLP. The underrepresentation of African languages in existing models creates systems that serve their speakers poorly. By evaluating what currently works and what does not, this study contributes toward a more inclusive research agenda.

2 Background and Related Work

2.1 Semantic Textual Relatedness

Semantic textual relatedness differs from semantic similarity in important ways. Relatedness is broader: two sentences may be highly related without being similar. The SemRel dataset (Abdulmumin et al., 2024) was specifically designed to capture this broader notion using a comparative annotation framework that improves inter-annotator agreement and produces continuous relatedness scores between 0 and 1.

2.2 Multilingual Pretrained Models

XLM-R (Conneau et al., 2020), trained on 100 languages using masked language modelling over Common Crawl, achieves strong cross-lingual transfer. However, performance on low-resource African languages is constrained by their small proportion of pretraining data. AfriBERTa (Alabi et al., 2022) was pretrained on eleven African languages including Hausa and several Bantu languages, providing more culturally grounded representations. Sentence-BERT (Reimers and Gurevych, 2019) enables efficient computation of semantic similarity through fixed-size sentence embeddings.

2.3 Data Augmentation for Low-Resource NLP

Back-translation (Sennrich et al., 2016) produces paraphrase-like variants by translating sentences to another language and back, increasing training diversity. For African languages, translation quality varies substantially; Hausa has moderate coverage through multilingual models such as Helsinki-NLP opus-mt-mul-en (Helsinki-NLP, 2020), while Kinyarwanda has more limited support.

Despite these developments, there is limited work performing targeted cross-lingual transfer analysis across multiple African languages using SemRel. This project addresses this gap through a systematic comparative study.

3 Methodology

3.1 Dataset

The primary dataset is SemRel (Abdulmumin et al., 2024), a multilingual collection of sentence pairs annotated with continuous relatedness scores between 0 and 1. Table 1 shows the split sizes used. Afrikaans has no training split, constraining it to zero-shot evaluation only. Score distributions are approximately uniform and centred around 0.5 across all languages (mean range: 0.495–0.517), confirming a well-balanced dataset. Sentence lengths are shortest for English (mean 12.5 words, p95 = 22) and Afrikaans (mean 14.0, p95 = 18), and longer for Hausa (mean 19.3, p95 = 25) and Kinyarwanda (mean 20.2, p95 = 25), all within the 128-token fine-tuning limit.

Language	Train	Dev	Test
English	5,500	250	2,600
Afrikaans	–	375	375
Hausa	1,736	212	603
Kinyarwanda	778	102	222

Table 1: SemRel dataset split sizes per language.

3.2 Models

Three modelling approaches are evaluated:

Baseline. paraphrase-multilingual-mpnet-base-v2 (Reimers and Gurevych, 2019) generates fixed-size sentence embeddings; cosine similarity is used directly as the relatedness score without fine-tuning.

XLM-R. XLM-R-base (Conneau et al., 2020) is fine-tuned for regression by adding a linear projection head over the [CLS] token. The model is first fine-tuned on English, then evaluated zero-shot on African languages. Where training data is available (Hausa, Kinyarwanda), XLM-R is additionally fine-tuned on those splits.

AfriBERTa. AfriBERTa-large (Alabi et al., 2022) is fine-tuned following the same procedure as XLM-R. Comparison between the two isolates the effect of African-language pretraining.

3.3 Training Setup

All fine-tuning uses MSE loss, AdamW optimiser ($\text{lr} = 2\text{e-}5$, weight decay = 0.01), linear warmup over 10% of training steps, maximum 10 epochs, early stopping with patience 3 on dev Spearman ρ . Batch size is 16 for XLM-R and 32 for AfriBERTa. All experiments run on Kaggle T4 GPU. All models are implemented using the Transformers library (Wolf et al., 2020).

3.4 Data Augmentation

Back-translation augmentation is applied to Hausa and Kinyarwanda training sets using Helsinki-NLP/opus-mt-mul-en and opus-mt-en-mul (Helsinki-NLP, 2020). Each sentence is translated to English and back to the source language. Quality filtering discards pairs where either back-translated sentence has cosine similarity below a threshold with the original embedding. Augmented data is concatenated with the original training set for retraining. Results and quality statistics are reported in Section 4.6.

3.5 Evaluation

The primary metric is Spearman rank correlation coefficient (ρ). MSE is used as a secondary metric. All evaluation is on held-out test sets.

4 Experiments and Results

4.1 Main Results

Table 2 presents results for all experiments. XLM-R outperforms AfriBERTa in every condition. Bold indicates the best result per language.

4.2 Baseline Performance

The cosine similarity baseline achieves $\rho = 0.805$ on English and $\rho = 0.765$ on Afrikaans, both substantially above 0.5, confirming that off-the-shelf multilingual embeddings capture semantic relatedness well for these Germanic languages. Performance drops sharply for Hausa ($\rho = 0.318$, MSE = 0.135) and Kinyarwanda ($\rho = 0.362$, MSE = 0.143), with MSE approximately four times higher than English (MSE = 0.035), establishing a clear motivation for fine-tuning.

4.3 Cross-Lingual Transfer

Zero-shot evaluation of the English-fine-tuned XLM-R demonstrates substantial improvement over the baseline across all languages. For Afrikaans, zero-shot transfer achieves $\rho = 0.818$,

remarkably exceeding the English fine-tuned model’s performance on English itself ($\rho = 0.835$), suggesting near-perfect transfer due to Germanic language proximity. This directly addresses Research Question 1. For Hausa, zero-shot achieves $\rho = 0.642$ (+0.325 over baseline), and for Kinyarwanda $\rho = 0.482$ (+0.120), the smallest gain reflecting greater linguistic distance from English.

4.4 Language-Specific Fine-Tuning

Fine-tuning on language-specific data yields further improvements. For Hausa, fine-tuning achieves $\rho = 0.686$ vs $\rho = 0.642$ zero-shot (+0.044). For Kinyarwanda, fine-tuning achieves $\rho = 0.634$ vs $\rho = 0.482$ zero-shot (+0.152), the largest gain among all conditions, confirming that Kinyarwanda requires in-language data to a greater degree than the other languages. The training curve for Kinyarwanda began near zero (dev $\rho = 0.080$ at epoch 1) and climbed steadily, reflecting initially poor representations.

4.5 Multilingual vs. African-Focused Models

Contrary to our hypothesis, XLM-R outperforms AfriBERTa in every experimental condition, addressing Research Question 2. In the fine-tuned setting, XLM-R achieves $\rho = 0.686$ vs AfriBERTa’s $\rho = 0.476$ on Hausa, and $\rho = 0.634$ vs $\rho = 0.493$ on Kinyarwanda. We attribute this to XLM-R’s substantially larger pretraining corpus and vocabulary: for semantic relatedness regression, scale of multilingual pretraining outweighs language-specific focus at the model sizes evaluated.

4.6 Data Augmentation

Back-translation using Helsinki-NLP/opus-mt-mul-en produced augmented training sets for Hausa and Kinyarwanda. For Hausa, a quality threshold of 0.75 yielded only 0.2% survival (3 of 1,736 pairs), confirming limited translation resource maturity. Lowering the threshold to 0.30 produced 1,081 augmented pairs (62.3% survival), increasing the Hausa training set from 1,736 to 2,817 pairs. For Kinyarwanda, back-translation at threshold 0.30 produced 777 pairs (99.9% survival), increasing the training set from 778 to 1,555 pairs.

XLM-R retrained on augmented Hausa achieves $\rho = 0.698$ (AU-1), a meaningful improvement over the non-augmented result ($\rho = 0.686$, TL-2). For Kinyarwanda, augmented XLM-R achieves

Exp.	Model	Language	ρ	MSE	Aug.
BL-1	Cosine baseline	English	0.8048	0.0352	No
BL-1	Cosine baseline	Afrikaans	0.7646	0.0263	No
BL-1	Cosine baseline	Hausa	0.3175	0.1346	No
BL-1	Cosine baseline	Kinyarwanda	0.3617	0.1426	No
TL-ENG	XLM-R fine-tuned	English	0.8353	0.0200	No
TL-1	XLM-R zero-shot	Afrikaans	0.8177	0.0159	No
TL-1	XLM-R zero-shot	Hausa	0.6422	0.0539	No
TL-1	XLM-R zero-shot	Kinyarwanda	0.4819	0.0349	No
TL-2	XLM-R fine-tuned	Hausa	0.6862	0.0404	No
TL-2	XLM-R fine-tuned	Kinyarwanda	0.6339	0.0269	No
TL-3	AfriBERTa zero-shot	Afrikaans	0.5354	0.1181	No
TL-3	AfriBERTa zero-shot	Hausa	0.3108	0.0724	No
TL-3	AfriBERTa zero-shot	Kinyarwanda	0.3524	0.0496	No
TL-4	AfriBERTa fine-tuned	Hausa	0.4756	0.0807	No
TL-4	AfriBERTa fine-tuned	Kinyarwanda	0.4926	0.0366	No
AU-1	XLM-R + aug.	Hausa	0.6984	0.0470	Yes
AU-2	AfriBERTa + aug.	Hausa	0.4776	0.0606	Yes
AU-3	XLM-R + aug.	Kinyarwanda	0.6432	0.0289	Yes
AU-4	AfriBERTa + aug.	Kinyarwanda	0.3951	0.0370	Yes

Table 2: Spearman ρ and MSE for all experiments on test sets. Bold = best per language.

$\rho = 0.643$ (AU-3), also improving over the non-augmented result ($\rho = 0.634$). AfriBERTa with augmentation achieves $\rho = 0.478$ on Hausa (AU-2) and $\rho = 0.395$ on Kinyarwanda (AU-4), consistent with AfriBERTa’s weaker overall performance. These results suggest that back-translation augmentation provides modest but consistent improvements for XLM-R across both languages, directly addressing Research Question 3. The gains are more pronounced for Hausa (+0.012) than Kinyarwanda (+0.009), despite Hausa having lower translation quality, likely because the larger augmented set (1,081 pairs vs 777) provides more training signal.

4.7 Error Analysis

Error analysis on the ten highest-error test pairs per language reveals three recurring patterns. For Hausa, domain-specific vocabulary (political, agricultural, judicial) is consistently underestimated, reflecting limited domain coverage in pretraining corpora. For Kinyarwanda, errors concentrate in verb-heavy constructions consistent with agglutinative Bantu morphology, which subword tokenisation handles poorly. For Afrikaans, the model occasionally over-predicts relatedness for code-switching and social media language underrepresented in SemRel.

5 Reflections and Discussion

5.1 What Worked

The overall pipeline proved replicable using Hugging Face transformers and the SemRel dataset. The most consistent finding is that even small amounts of in-language labelled data combined with a strong pretrained model produces meaningful gains over zero-shot baselines. The unexpected superiority of XLM-R over AfriBERTa provides important empirical evidence that scale of pretraining data outweighs language-specific focus for this task.

5.2 Challenges and Limitations

The most significant challenge was the absence of a training split for Afrikaans in SemRel, which constrained evaluation to zero-shot cross-lingual transfer only and limits direct comparison with Hausa and Kinyarwanda results. Future work could investigate semi-supervised approaches or targeted data collection for Afrikaans.

Back-translation quality for Hausa was severely limited by the multilingual translation model (opus-mt-mul-en), with only 0.2% of pairs surviving a cosine similarity threshold of 0.75. Even at a lowered threshold of 0.30, augmentation produced modest performance gains (+0.012 for XLM-R on Hausa, $\rho = 0.698$ vs 0.686; +0.009 for Kinyarwanda, $\rho = 0.643$ vs 0.634). The stark contrast with Kinyarwanda, which achieved a 99.9% survival rate at the same threshold, highlights substan-

tial variation in translation resource quality even within low-resource African languages. The Kinyarwanda augmented result ($\rho = 0.643$) shows a small improvement over the non-augmented result ($\rho = 0.634$), suggesting that the augmented pairs, while syntactically valid, may not provide sufficiently diverse training signal at this quality level. Future work should explore dedicated language-specific translation models or alternative augmentation strategies such as synonym substitution or cross-lingual paraphrase generation.

The unexpected superiority of XLM-R over AfriBERTa across all conditions warrants further investigation. We attribute this to XLM-R’s substantially larger pretraining corpus (2.5TB vs AfriBERTa’s significantly smaller African language corpora), but evaluation of larger AfriBERTa variants or more recent African-focused models such as AfroXLMR-large may yield different conclusions.

5.3 Responsible NLP

Language diversity: This study explicitly targets three underrepresented languages, though coverage within African NLP remains uneven.

Data ethics: SemRel is openly available for academic research. Annotation quality and compensations are not uniformly documented across languages, raising questions about consistency and fair labour practices.

Equity and bias: Performance gaps between English/Afrikaans and Hausa/Kinyarwanda reflect structural inequities in training data availability, not properties of the languages themselves. Systems deployed based on these models would provide worse service to Hausa and Kinyarwanda speakers. *Societal impact:* Improved semantic relatedness models for African languages have potential to benefit information retrieval and educational technology across the continent. However, current Kinyarwanda performance ($\rho \approx 0.63$) remains insufficient for high-stakes deployment.

6 Conclusion

This study evaluated cross-lingual transfer and data augmentation for semantic relatedness in Afrikaans, Hausa, and Kinyarwanda using SemRel. Three key findings emerge. First, XLM-R zero-shot transfer is highly effective for Afrikaans ($\rho = 0.818$) but weaker for Hausa and Kinyarwanda, where language-specific fine-tuning provides gains of +0.044 and +0.152 respectively. Second, XLM-

R consistently outperforms AfriBERTa in all conditions, indicating that pretraining scale outweighs language-specific focus for this task. Third, data augmentation via back-translation provides modest improvements for both languages (Hausa: $\rho = 0.698$ vs 0.686; Kinyarwanda: $\rho = 0.643$ vs 0.634 for XLM-R), with gains constrained by translation quality and the relatively small size of the augmented training sets.

Limitations

This study evaluates only two model families (XLM-R-base and AfriBERTa-large) and does not include larger language models or more recent African-focused models such as AfroXLMR. Results are limited to the SemRel dataset and may not generalise to other semantic relatedness benchmarks or domains. The absence of an Afrikaans training split prevents direct comparison of fine-tuning performance across all three target languages. Back-translation augmentation was constrained by the quality of available multilingual translation models; dedicated Hausa–English translation resources may yield substantially different augmentation outcomes. All experiments were conducted on a single T4 GPU with limited computational budget, precluding hyperparameter search or ensemble methods that might improve performance.

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